

IS 733 - Lab 11

Part I: Distributional Hypothesis

Please use the Distributional Hypothesis Concept to find the solution, show your work step by step

Answer:

Part I – Distributional Hypothesis

The Distributional Hypothesis means that words that appear in similar situations often have similar meanings. Using this idea, I looked at the words that best fit into each group of sentences.

In **Example 1**, the sentences included phrases like “a piece of,” “eating,” “cut with a knife,” and “made from milk.” These phrases are commonly used with the word **cheese**. The other words, like “cake” and “butter,” don’t match all the situations. For example, cake isn’t made from just milk, and I don’t usually say “a piece of butter.”

In **Example 2**, the correct word is **car** because it fits all the sentence clues like “parked in driveway,” “bought,” “drive fast,” and “wash on weekends.” These are things people often do with cars. While trucks and motorcycles might also fit some of these, they don’t fit all of them as well as car does.

In **Example 3**, the best match is **book**. Words like “read,” “enjoy before bed,” “has chapters and a cover,” and “borrow from the library” are all things I normally say about books. While “story” or “novel” are close, “book” fits all the sentences more clearly.

Part – II is attached as a python notebook.